

POLITSC 3780: Data Literacy and Visualization

The Ohio State University, Autumn (2023)

Instructor: Liuya Zhang

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In-Person Recitation: Monday 10:20 - 11:15 (Derby Hall 125)

Live Zoom Recitation: Monday 1:00 - 2:00

Recitation Zoom Info:

Office Hours: Wednesday 4:00 - 5:00 or by appointment (Derby 2012)

OH Meeting Info:

TA: Ryan Dawe

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Office Hours See Carmen

Course Description

Many, if not most, of the major debates in modern political science revolve around questions that can be addressed with data. The sources of voting behavior, the correlates of war, the determinants of development, political economy, psychology, institutions, and conflict -all are issues that are amenable to data-based analysis.

At the same time, the amount of available data and the number of publicly-available open-source tools for cleaning, transforming, analyzing and visualizing it have increased exponentially since the turn of the millennium. With a few clicks students can compare word frequencies in books over time or construct elaborate size-weighted wordclouds -tasks that would have taken scholars weeks if not months of effort in the past.

This course introduces students to those tools and the principles behind their use in the context of applications in political science. It marries the substance of political theory to the methodologies of data visualization and exploratory data analysis. It neither requires nor imparts any statistical background: it is designed to serve either as a standalone course or as a gateway to a more advanced data-analytics class.

GE Goals and Expected Learning Outcomes

Goals

***Students** develop skills in drawing conclusions and critically evaluating results based on data.*

***Students** will gain basic knowledge across the major subfields and thematic specializations of Political Science.*

***Students** learn visualization techniques for the presentation of different types of data.*

Achieving Expected Learning Outcomes

Students understand basic concepts of statistics and probability, comprehend methods needed to analyze and critically evaluate statistical arguments, and recognize the importance of statistical ideas.

The course will fulfill the General Education (GE) requirement in Data Analysis by helping students develop skills in drawing conclusions and critically evaluating arguments based on data. It will introduce students to basic concepts in statistics and probability and focus heavily on data visualization. It will tie these elements together with the logic of research design in order to give students the ability to evaluate statistical arguments, and it will show them examples of how to do so using data on such topics as American partisanship and elections, public opinion, and international conflicts.

Assessment of Expected Learning Outcomes

The effectiveness of this course in achieving the expected learning outcomes outlined above will be determined in three ways.

1. A short problem set will be assigned weekly to help students internalize the material provided in the online lectures and reviewed in recitation. The problem sets will be graded and students' mastery of the relevant skills will be assessed.
2. During the class, students will work on an independent project of their choosing, one that involves answering questions using the skills they have acquired. This final assignment will count for 40% of the course grade.
3. Students will be urged to fill in the online Student Evaluation of Instruction (SEI) reports to assess the amount that they have learned in this course relative to others.

Materials

Video Lectures

Recorded lectures are online and can be accessed via YouTube [here](#). These lectures should be watched before recitation.

The lectures are recorded in Quicktime format, which should be viewable on nearly any computer or mobile device. One advantage to the format is that, on most devices, lectures can be sped up to 1.4x to 1.5x; beyond about 2x, most of them become unintelligible. Students are advised to take advantage of this feature, as humans can generally understand speech at a higher rate than they can produce it. Be advised, though, that higher speeds generally require more focused attention, as important details are easier to miss. It's also possible to rewind and slow the recording back down if a particular section moves quickly or is difficult to understand.

Also, note that this course was originally developed as a variant of Political Science 4781, and most of the lectures were recorded before it had its new number, so do not be surprised to hear the wrong course number at the beginning of each lecture.

Recitation Recordings

Online recitations will be recorded for those unable to attend synchronously. You can choose between attending online or in-person recitations, or access the recorded recitation sessions if you are unable to attend.

The online lectures provide an introduction to concepts or teach a particular approach in R. Recitation will reinforce concepts and/or teach an additional approach in R. Especially in the second half of the semester, students will be exposed to several options for creating their own visualizations in R. Any techniques are acceptable for assignments and the final project.

Office Hours

Office hours held throughout the week are open times for students to work on problem sets and ask course-related questions. Students are expected to have watched the lectures and recitations before asking questions in office hours.

Recommended Texts

No purchases are expected for this course, only free resources are used. Two books are recommended for students that want additional reference, and they are available either as a Kindle book, which can be read on Kindle tablet or PC apps, or as an online resource (see links below). Reminder: you do not likely need to purchase the texts, as online resource should suffice.

1. Healy, Kieran. 2018. *Data Visualization: A Practical Introduction*. Princeton: Princeton University Press. (Most of the book and lecture notes are available [free online](#))
2. Wickham, Hadley and Garrett Grolemund. 2017. *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data*. Sebastopol, California: O'Reilly Media. (Most of the book and lecture notes are available [free online](#))

Software

In this course, we will use the statistical software called R(see <https://www.r-project.org>). We will also use RStudio(see <https://www.rstudio.com>), a text editor that will make it easy for you to interact with R. Both R and RStudio are free, and you should download each on your own computer (Mac or PC). The software can also be accessed remotely using [t](#)hrough some procedures are slightly different.

What is R and why use it?

- Widely-used in academia and industries
- Open-source and free
- Power and flexibility
- Graphical capabilities
- Learning R = learning basic programming

Resources for Using R

Students invariably find learning R to be the most challenging part of this course. It is also, in the long run, the most rewarding, in that it's a valuable and marketable skill that also prepares you for more advanced classes. The learning curve is very steep, however, and it is best to be prepared. Toward that end, students should take advantage of the following:

- *Attend recitation.* Although attendance is not officially monitored, the best way to succeed in this course is by coming to the weekly recitation sessions. During these sessions, your instructor will give helpful tips for completing assignments, and will be available to go over material and answer any questions about the lectures and homework. There are 3 hour-long sections each week that you can choose to attend. If you have really been struggling with a problem set, you can even attend more than one session per week. Take advantage of these times to work on assignments and get help with course material.
- *Monitor Carmen Discussions and Announcements.* These sources are the communications hubs for the course. You can go here to post questions about assignments, and where you can answer questions as well. If you have a question about an assignment, lecture, or how to use R, chances are that others will have the same question. In many instances, you will find that your question has already been asked (and answered!) by your peers. Note that the instructor may not be monitoring the discussions everyday.
- *Google is your friend.* Simply google your coding question, such as “how do I make a bar chart in R,” and you are bound to find many useful resources on the web.
- *Use ChatGPT.* You can utilize ChatGPT for coding help in this course. Keep in mind that ChatGPT's information is current up until September 2021. Be cautious of potential errors and outdated details. While it's a useful resource, always cross-reference with recent R updates and exercise your judgment. Remember, ChatGPT can provide guidance, but your critical thinking and due diligence are crucial for accurate results.
- *Statistical consulting at PRISM.* The Department of Political Science offers help through the [Program In Statistics and Methodology](#)(PRISM). You may contact the PRISM Fellows to setup a brief appointment for assistance.
- *Online tutorials and books.* You should download Shidan Murphy's excellent [R Instructor app](#)for iPhone or Android and bookmark, or just download and print out, Tom Short's fantastic [R Reference Card](#). You may also wish to consider Paul Teetor's [R Cookbook](#)from O'Reilly. You might also consider the following book: Hrishi, Mittal. [R Graph Cookbook](#)(Kindle edition; Packt Publishing)

Course requirements

There will not be a traditional midterm or final exam. Rather, weekly of assignment will make up 60% of the grade, and the remaining 40% will come from a final project in which the student finds a dataset in his or her area of interest that is not already used in the course, analyzes it to assess the structure of the data, and works through the most appropriate, succinct, and informative summaries and visualizations. The final course grade will be determined based on the following breakdown and grading scale:

Problem Sets	60%	B+	87–89%	A	93–100%	A–	90–92%
Final Project	40%	C+	77–79%	B	84–86%	B–	80–83%
		D+	67–69%	C	74–76%	C–	70–73%
		E	< 60%	D	64–66%	D–	60–63%

Problem Sets

Assignments will be posted on Carmen at least two weeks in advance of their Friday due date; assignments will be **due Friday by 5:00 PM**. Assignments can be completed in groups of up to three students. The grade assigned to a group will be assigned to each individual in the group. The final project will be an individual project, and will be based on many of the skills obtained throughout the semester, so there is little sense in joining a group and not being an active contributor. Students who submit a 1-page description of their research design for the final project in Week 6 will receive **3 Bonus points**. The guidelines for the final project will be available on Carmen on September 25.

Final Project

The final project is an **individual** project. The final project is designed to highlight your proficiency in creating visualizations to help address a political theory with data. The final project is a brief research paper with particular attention paid to supporting visualizations. As long as all professors agree, the final project in this course may be used to fill requirements of other courses taken concurrently.

Policies and procedures

Communication

Outside of class, Carmen will be the preferred methods of contact. Announcements will be made via Carmen and the Discussion page will hold questions, tips, and pointers. *Your University issued email will be used, so please be sure that you have access to that account and that you check it regularly.* I will try to answer any emails within 24 hours during the week, and answer weekend emails on Monday, so please prepare to write to me well in advance if you have questions about the course. The instructor will make it a priority to respond as quickly as possible to emergencies and other extreme issues. In addition, please make use of online office hours, as that is the time allocated to be 100% available to you. If you have any questions or are having difficulty completing course requirements, please contact me as soon as possible.

Accommodations

I encourage you to begin assignments well ahead of time, as everyone is aware of the heavy workload you will be facing with other courses. Please be aware that I will not make any concessions regarding workload (such as dispensing when you miss class or excusing you from having done assignments). You are responsible for distributing your time according to the time you have every week. I am, however, well aware of the potential negative effect of exogenous factors on your learning or the possibility that you may be facing a tough time or an illness. I will require written confirmation from the counsellor or your physician to engage in dispensation and accommodation, but I will be

more than happy to help in any way I can, within reasonable limits. Specific policies for assignments and the final project are below.

Assignment dispensation policy

Assignments must be submitted on Carmen *on the day they are due (by 5 PM)*. If a student is unable to complete an assignment, they will be allowed to turn it in late only if the absence is due to a *documented* medical, family, or similar serious emergency, observance of religious holy days (which requires written notification to the instructor at least 14 days prior to the requested absence date), or properly documented University-sponsored planned activities. *Incomplete assignments in all other cases will result in a score of zero*. If you become aware that you will not be able to complete an assignment or final project ahead of time, please contact the instructor and seek permission for an extension as soon as possible.

Academic misconduct

It is the responsibility of the Committee on Academic Misconduct to investigate or establish procedures for the investigation of all reported cases of student academic misconduct. The term “academic misconduct” includes all forms of student academic misconduct wherever committed; illustrated by, but not limited to, cases of plagiarism and dishonest practices in connection with examinations. The instructor shall report all instances of alleged academic misconduct to the committee (Faculty Rule 3335–5–487). For additional information, see the Code of Student Conduct (http://studentaffairs.osu.edu/resource_csc.asp).

Disability services

Students with disabilities that have been certified by the Office for Disability Services will be appropriately accommodated, and should inform the instructor as soon as possible of their needs. The Office for Disability Services is located in 150 Pomerene Hall, 1760 Neil Avenue; telephone 292–3307, TDD 292–0901; <http://www.ods.ohio-state.edu>.

Lecture Schedule

Week	Subject	Week Start	Assignment - Friday 5PM
Week 1 Introduction	Syllabus and policy	8/28	
Week 2 Labor Day		9/4 (No class)	
Week 3 Web-scraping and Data tools	Lectures 1, 2, 3	9/11	Assignment 1 9/15
Week 4 Guideline of Visualization	Lectures 4, 5, 9	9/18	Assignment 2 (9/22)
Week 5 Spatial and Network Data Visualization	Lectures 6, 7	9/25	Assignment 3 (9/29)
Week 6 Research Design	Lecture 17, 20	10/2	Bonus 3 (10/6)
Week 7 Introduction to R	Lectures 8	10/9	
Week 8 Data structures and transformation	Lecture 10	10/16	Assignment 4 (10/20)
Week 9 Basic R Plots I	Lecture 11	10/23	Assignment 5 (10/27)
Week 10 Basic R Plots II	Lecture 12	10/30	Assignment 6 (11/3)
Week 11 Simulations and Uncertainty	Lecture 19, 21	11/6	
Week 12 APIs	Lecture 13	11/13	Assignment 7 (11/17)
Week 13 Advanced R plots I - Spatial data	Lectures 14, 16	11/20	Assignment 8 (11/24)
Week 14 Advanced R plots II - Text data (1)	Lectures 15, 18	11/27	Assignment 9 (12/8)
Week 15 Advanced R plots III - Text data (2)		12/4	*Final Project (12/15)*

Provided that recitation sessions aim to help students digest material and work on assignments, students are expected to listen to and view all the corresponding video lectures prior to each session.

Description of Lectures

Lecture 1: Introduction. Nature of the course, its relationship to other courses and students' plans of study; how and why data visualization can be useful.

Lecture 2: Data are All Around Us! Terminology, sources, data collection and cleaning (OpenRefine)

Lecture 3: Data Tools, part I. Open-source solutions that require little or no additional information or effort to produce compelling results.

Lecture 4: The Good. Exemplary representations of data from our field and others, with a discussion of what makes them so compelling.

Lecture 5: The Bad and The Ugly. Ideas that seemed good at the time, why they really aren't, how we can fix them, and what general principles they imply.

Lecture 6: Data Tools, part II. These versatile tools allow you to upload, explore, and visualize your own data.

Lecture 7: Truth. Data visualization is inherently a reduction of information. The main imperative, in reducing information, is integrity: representing the data as honestly as possible. We look at some good cases and some not-so-good cases.

Lecture 8: Introduction to R. A powerful (and free) general statistical package, R is capable of producing a wide range of graphs. In these lectures we demonstrate how to download and install the package, add libraries, import data, and create visualizations.

Lecture 9: Beauty. The difference between a fairly good data visualization and an amazing one often lies in the application of a few straightforward graphical principles.

Lecture 10: R, part II. Data structures and data transformations.

Lecture 11: Time. Different ways of examining variables over time.

Lecture 12: Plots in R. Scatterplots, line graphs, boxplots, dot charts, and more.

Lecture 13: APIs. The application program interface: a wonderful way for people to make their data available to you.

Lecture 14: Maps. Less common, but more eye-catching, are ways of looking for patterns across space, especially in maps.

Lecture 15: Careers in Data I. Aaron Schwartz describes his work at Community Research Partners.

Lecture 16: Space and Time. Representing relationships and trends over both space and time.

Lecture 17: Research Design. Designs for causal inference using observational data, including natural experiments, regression discontinuity, and differences in differences.

Lecture 18: Careers in Data II. Jill Riepenhoff and Mike Wagner discuss their work at the Columbus *Dispatch*.

Lecture 19: Simulated Worlds Using simulation to predict the outcome of the 2012 Presidential election.

Lecture 20: Workshop—Partisanship. Is American politics more partisan than it was 50 years ago? If so, why?

Lecture 21: Simulations and Inference Monte Carlo techniques for understanding uncertainty.